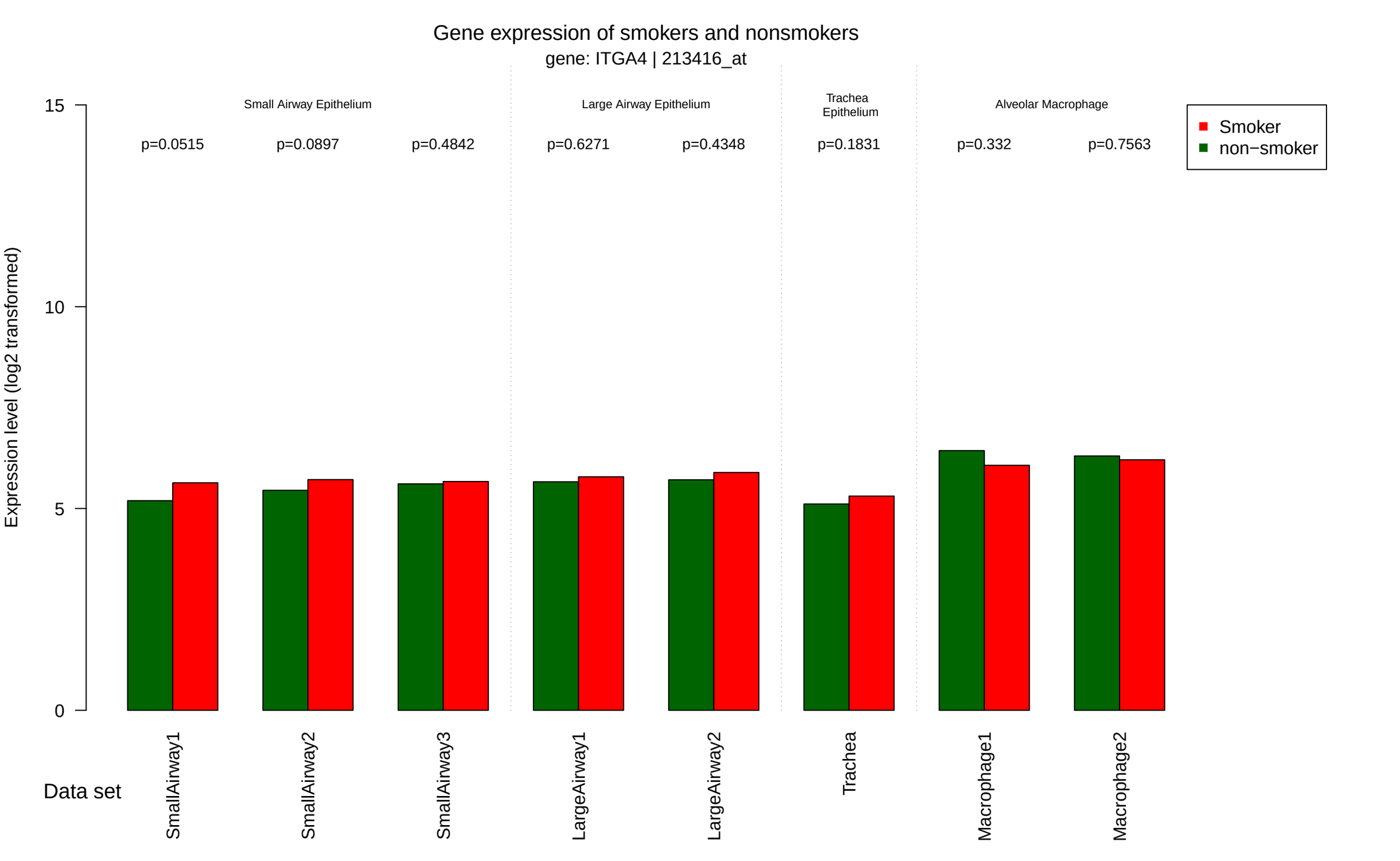
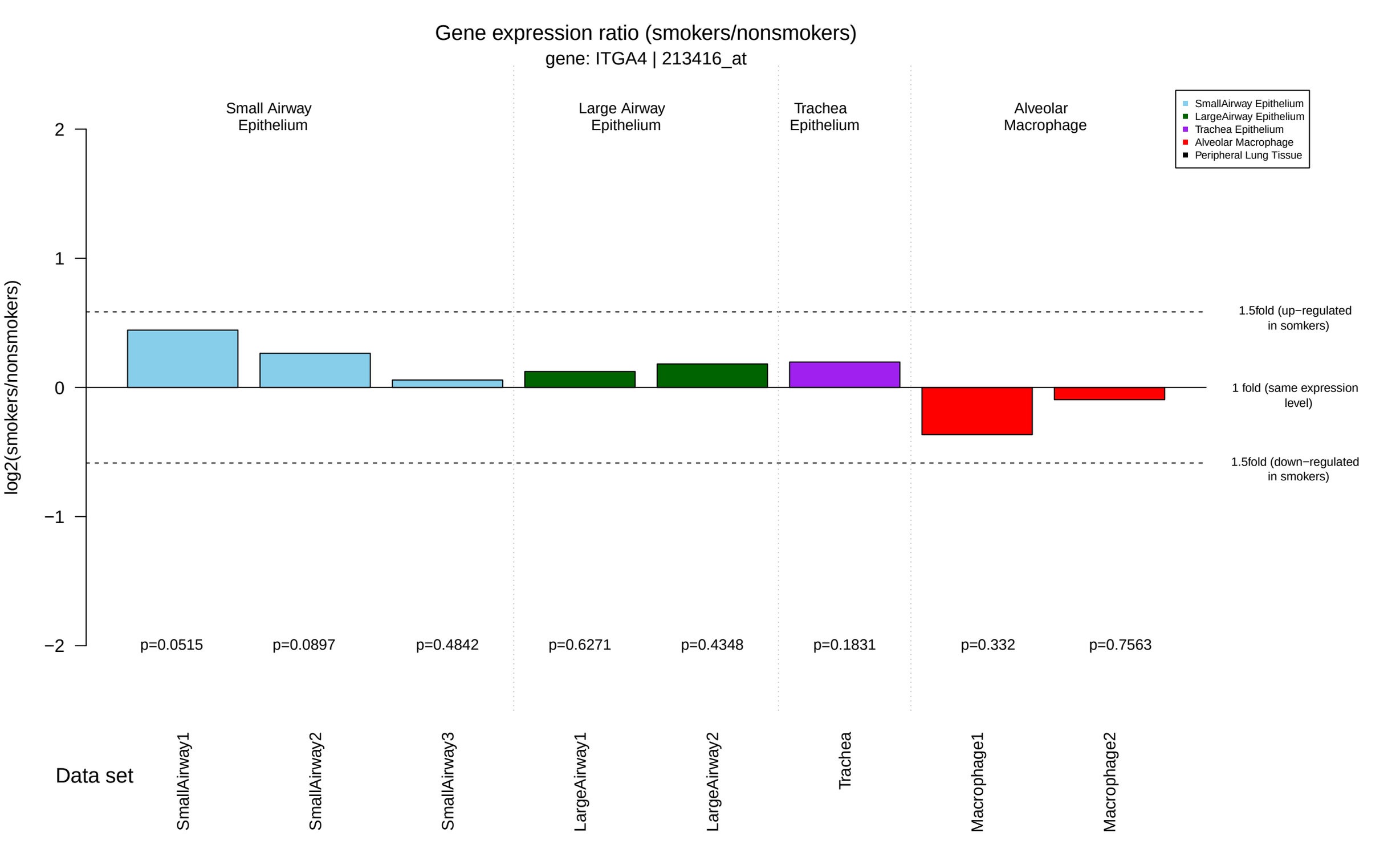
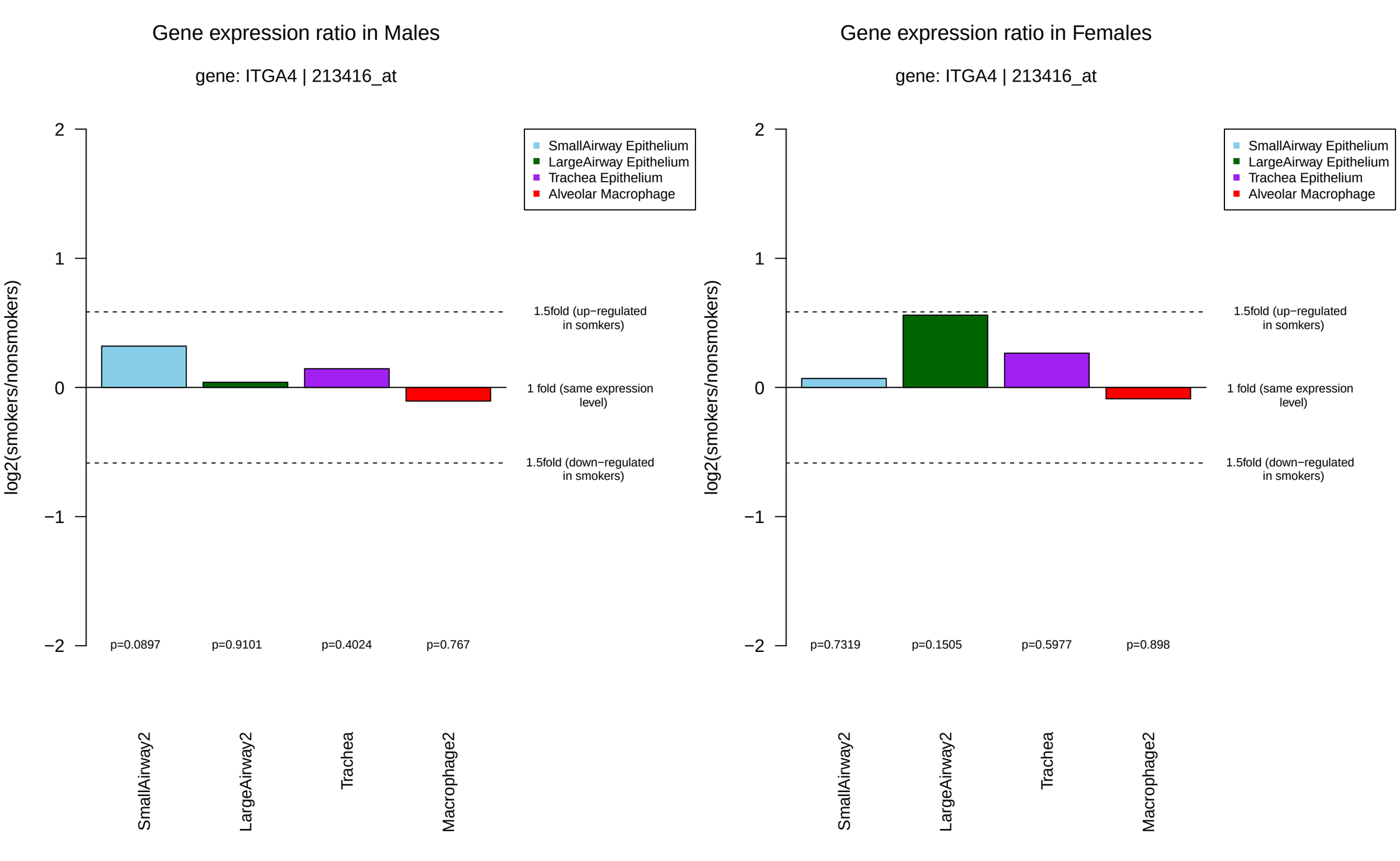
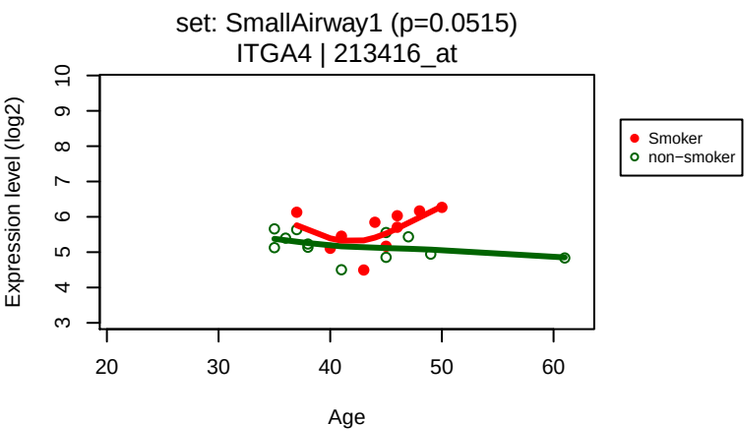
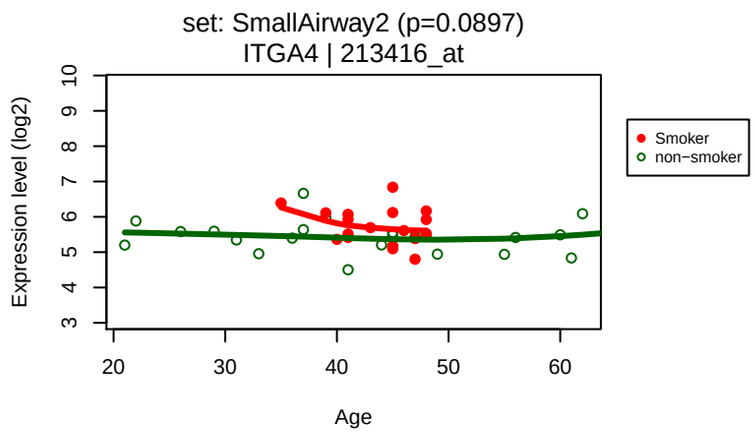
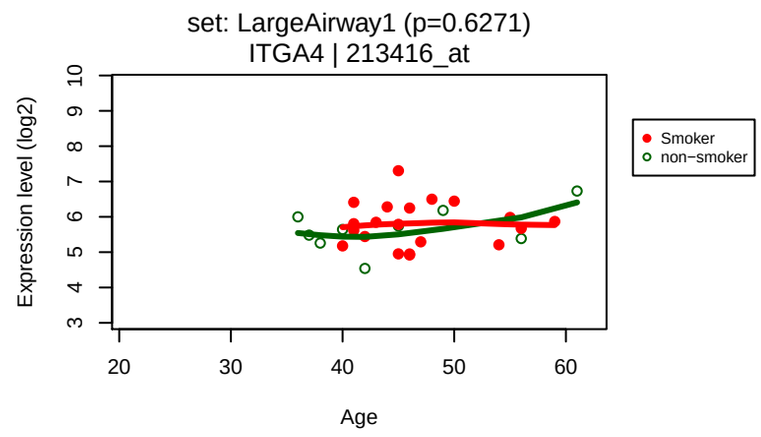

Gene expression of smokers and nonsmokers
gene: ITGA4 | 213416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0515
p=0.0897
p=0.4842
p=0.6271
p=0.4348
p=0.1831
p=0.332
p=0.7563
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

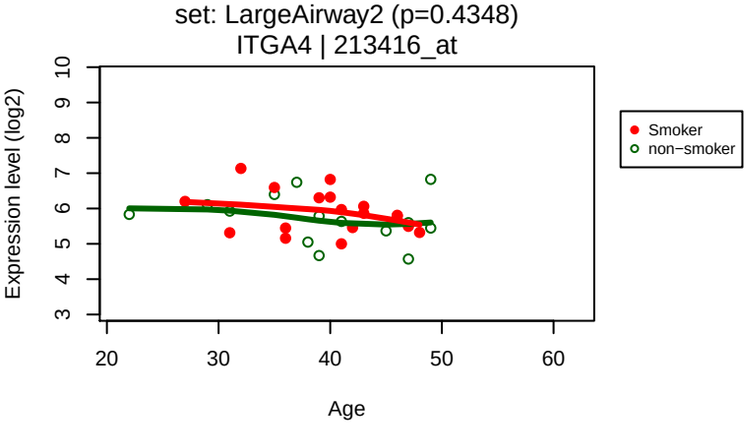

Gene expression ratio (smokers/nonsmokers)
gene: ITGA4 | 213416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0515
p=0.0897
p=0.4842
p=0.6271
p=0.4348
p=0.1831
p=0.332
p=0.7563
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

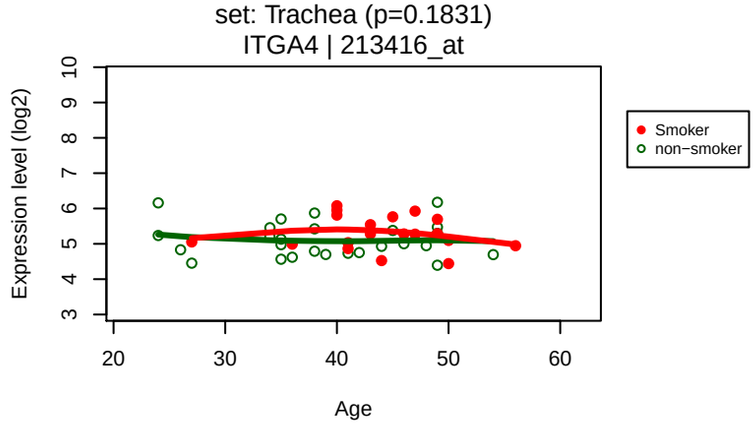

Gene expression ratio in Males
gene: ITGA4 | 213416_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0897
p=0.9101
p=0.4024
p=0.767
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITGA4 | 213416_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7319
p=0.1505
p=0.5977
p=0.898
SmallAirway2
LargeAirway2
Trachea
Macrophage2

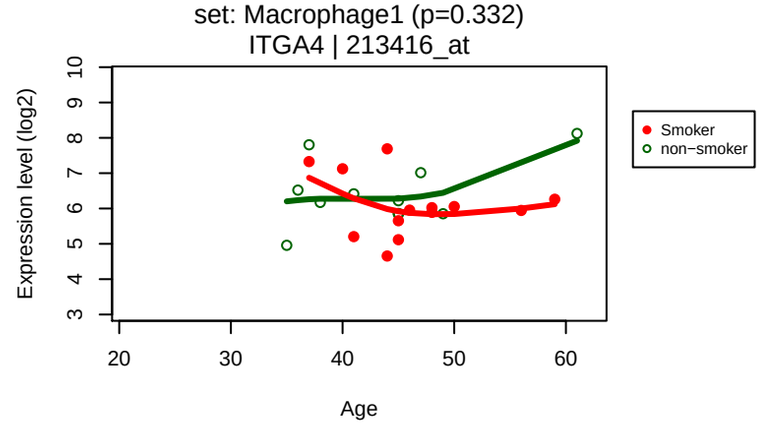

set: SmallAirway1 (p=0.0515)
ITGA4 | 213416_at
Expression level (log2)
Age
Smoker
non−smoker

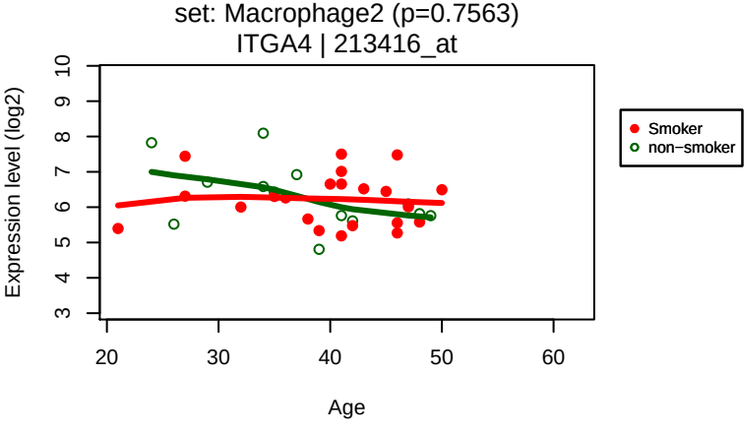

set: SmallAirway2 (p=0.0897)
ITGA4 | 213416_at
Expression level (log2)
Age
Smoker
non−smoker

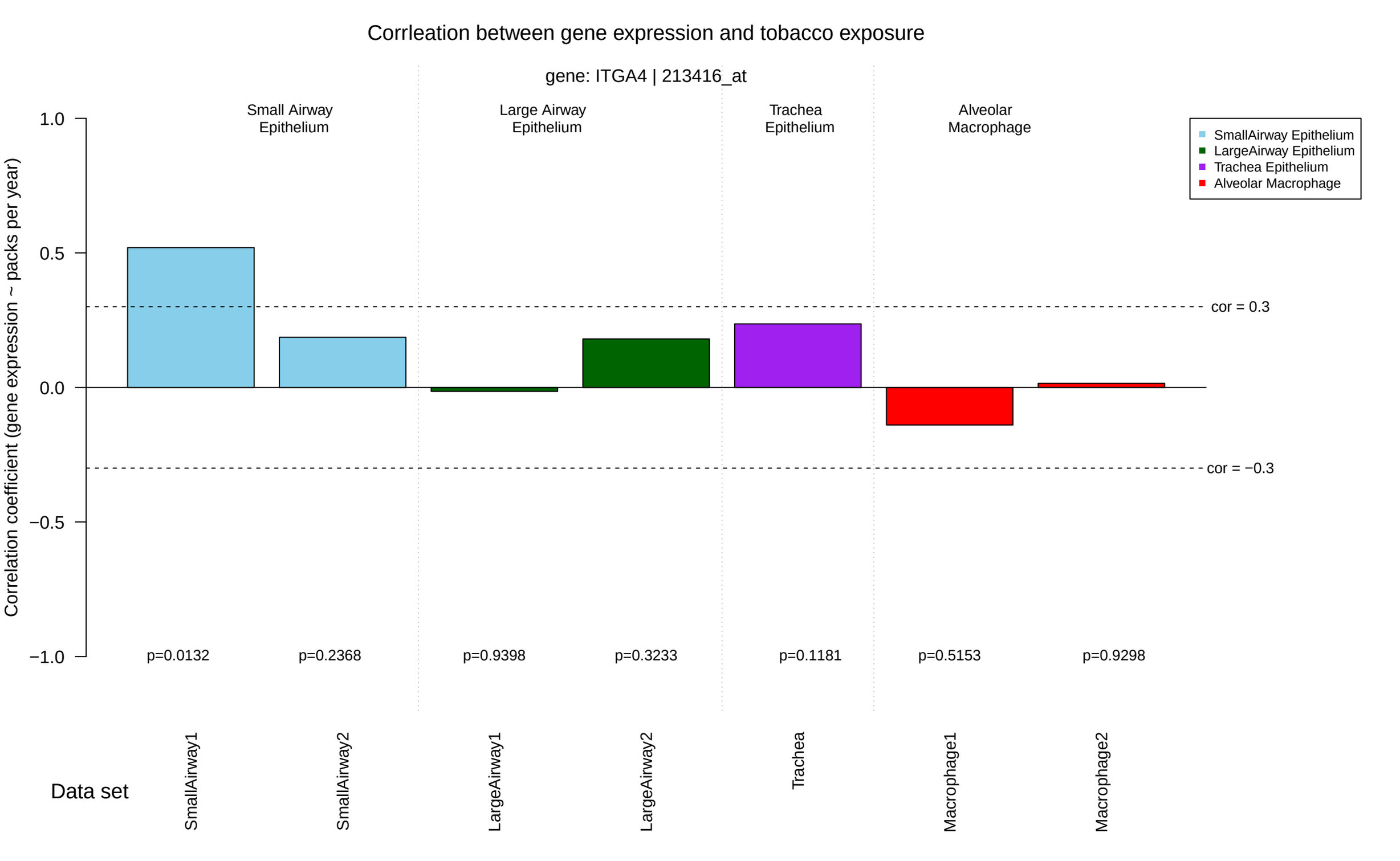

set: LargeAirway1 (p=0.6271)
ITGA4 | 213416_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4348)
ITGA4 | 213416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1831)
ITGA4 | 213416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.332)
ITGA4 | 213416_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7563)
ITGA4 | 213416_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITGA4 | 213416_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0132
p=0.2368
p=0.9398
p=0.3233
p=0.1181
p=0.5153
p=0.9298
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2